

Review of “Theorem 3: Cluster Consistency of State Discovery Under Strong Features”

Reviewer: Hostile reviewer for the Annals of Statistics **Manuscript:** Theorem 3 + Corollaries 1-3, SCX Theory **Decision:** **REJECT**

1. Overall Assessment

Recommendation: Reject. The proof contains at least two fatal mathematical errors that invalidate the main result. The algebra in the central calculation is wrong, the inequality directions are reversed in the critical step, and the derivation of the claimed $K = o(n^{1/3})$ rate is incoherent – the author walks through multiple self-contradictory calculations and never arrives at a consistent argument. Beyond these fatal errors, the proof relies on at least five unstated or unjustified assumptions, confuses the population parameter $\Delta_{\min}$ (a fixed property of the data) with an asymptotic scaling parameter, and never addresses the elephant in the room: k -means is NP-hard, but the proof requires the global empirical minimizer.

I do not recommend resubmission in the current form. The authors should either (a) restructure the proof around a tractable algorithm (e.g., Lloyd’s with a suitable initialization) or (b) weaken the claim to “there exists a polynomial-time algorithm that...” – and then prove it.

2. Critical Issues (Fatal If True)

Issue 1 (FATAL): The main misclassification bound diverges, not converges

This is the single most serious error. The algebra in the proof implies the opposite of the claimed result.

The proof establishes (lines 391-397):

$$\text{misclassification} \leq \frac{8\delta_n}{\pi_{\min}\Delta_{\min}^2}, \quad \delta_n = C_1 \sqrt{\frac{K \log n}{n}}$$

Under the separation condition $\Delta_{\min}^2 \geq C_0 \sigma^2 \frac{K \log n}{n}$, plugging the worst case (tightest bound, where $\Delta_{\min}^2 = \Theta(K \log n/n)$) gives:

$$\text{misclassification} \leq O\left(\frac{\sqrt{K \log n/n}}{K \log n/n}\right) = O\left(\sqrt{\frac{n}{K \log n}}\right)$$

The author writes in line 459 that this $\rightarrow 0$ requires $K \ll n/\log n$. **The inequality direction is reversed.** The quantity $\sqrt{n/(K \log n)} \rightarrow 0$ iff $K \log n$ grows faster than n , i.e., $K \gg n/\log n$. For any K that grows sublinearly in n (including $K = o(n^{1/3})$), $\sqrt{n/(K \log n)} \rightarrow \infty$. The bound explodes.

K scaling	$\sqrt{n/(K \log n)}$	Behavior
Fixed K	$\Theta(\sqrt{n})$	Diverges
$K = n^{1/3}$	$\Theta(n^{1/3}/\sqrt{\log n})$	Diverges
$K = n/\log n$	$\Theta(1)$	Constant
$K = n^{0.9}$	$\Theta(n^{0.05})$	Diverges (slowly)

K scaling	$\sqrt{n/(K \log n)}$	Behavior
$K = n/\sqrt{\log n}$	$\Theta(n^{-1/4})$	Converges

The only way to get convergence is $K = \omega(n/\log n)$, which is a *faster* growth rate – opposite to the claimed $o(n^{1/3})$, and would violate the separation condition (since centers would need to pack into a bounded region).

This is fatal. The theorem’s central claim – that misclassification $\rightarrow 0$ under $K = o(n^{1/3})$ – is mathematically impossible under the stated bounds.

Issue 2 (FATAL): The uniform convergence bound does not converge

In Step 2 of the proof of Theorem 3 (lines 362-377), the author sets:

$$\delta_n = C_1 \sqrt{\frac{K d_\phi \log n}{n}}$$

and claims:

$$\log \mathbb{P}(\sup |W_n - W| > \delta_n) \leq K d_\phi \log \left(1 + \frac{8C_1 M}{\delta_n} \right) - \frac{n\delta_n^2}{32C_1^2}$$

Substituting δ_n :

- First term: $K d_\phi \log \left(1 + \frac{8C_1 M}{C_1 \sqrt{K d_\phi \log n/n}} \right) = K d_\phi \log \left(1 + \frac{8M}{\sqrt{K d_\phi \log n/n}} \right)$
- For large n , $\sqrt{K d_\phi \log n/n} \rightarrow 0$, so the argument of log is $\sim \frac{8M}{\sqrt{K d_\phi \log n/n}}$, giving $\log(\cdot) \sim \frac{1}{2} \log n - \frac{1}{2} \log(K d_\phi \log n) + \text{constant}$

- So the first term is $\sim \frac{Kd_\phi}{2} \log n$ (up to lower-order terms).
- Second term: $\frac{n}{32C_1^2} \cdot C_1^2 \frac{Kd_\phi \log n}{n} = \frac{Kd_\phi \log n}{32}$

The total: $Kd_\phi \log n \left(\frac{1}{2} - \frac{1}{32}\right) + \text{lower order} = \frac{15}{32} Kd_\phi \log n + o(Kd_\phi \log n)$

This is positive, not negative. The claimed value of $Kd_\phi \log n - (Kd_\phi \log n)/32$ (line 371) would require the log term to equal $\log n$, which it does not – it equals $\frac{1}{2} \log n + o(\log n)$. The author made an algebraic mistake in simplifying the covering number term.

As $n \rightarrow \infty$, the probability bound tends to $+\infty$, meaning the bound is vacuous. The uniform convergence does not hold at this δ_n rate.

Issue 3 (FATAL): The proof assumes the global optimum of an NP-hard problem

Line 89 defines $\hat{\mu}_n^* = \arg \min W_n(\hat{\mu})$ – the global empirical minimizer. The proof never addresses that k -means is NP-hard (Aloise et al., 2009; Dasgupta, 2008). In practice, Lloyd’s algorithm finds a local optimum, and the gap between local and global minima can be arbitrarily large.

The author cites Pollard (1981), but Pollard’s strong consistency result is for the *global* k -means minimizer, which Pollard himself notes is a theoretical construct. Extending Pollard’s result to local optima is non-trivial – local minima can fail to be consistent even when the global minimum is (see e.g., Rakhlin & Caponnetto, 2007, who show that stability of Lloyd’s requires additional conditions).

The paper claims a result about the algorithm SCX uses (Lloyd’s / k -means), but proves a result about a computationally intractable oracle. This is a bridge too far.

Issue 4 (FATAL): Lemma 5’s core calculation is incorrect

The argument that $W(\mu) - W(\mu^*) = \sum_{k=1}^K \pi_k \cdot \delta_k^2$ (lines 277-298) is not mathematically justified. The author writes:

$$W(\mu) - W(\mu^*) = \sum_{k=1}^K \pi_k \cdot \mathbb{E} \left[\min_j \|\mu_k - \mu'_j + \varepsilon\|_2^2 - \|\varepsilon\|_2^2 \right]$$

then replaces $\min_j$ with $j^*(k) = \arg \min_j \|\mu_k - \mu'_j\|_2$, obtaining:

$$\mathbb{E} \left[\min_j \|\mu_k - \mu'_j + \varepsilon\|_2^2 \right] \geq \mathbb{E} \left[\|\mu_k - \mu'_{j^*(k)} + \varepsilon\|_2^2 \right]$$

and then computes this as $\delta_k^2 + \mathbb{E}[\|\varepsilon\|_2^2]$.

Three problems: 1. The inequality is reversed. Since $\min_j$ gives the *smallest* value, $\min_j \|a + \varepsilon\| \leq \|a_{j^*(k)} + \varepsilon\|$ for *any fixed* $j^*(k)$, not $\geq$. The author should have $\leq$ here. 2. Even with the correct direction, $j^*(k)$ depends on ε through the min – it is the index of the estimated center closest to $\mu_k + \varepsilon$, not the center closest to μ_k . These can differ when noise is large relative to separation. 3. The expectation of $\|\mu_k - \mu'_j + \varepsilon\|_2^2$ is $\|\mu_k - \mu'_j\|_2^2 + \mathbb{E}[\|\varepsilon\|_2^2]$ only when ε is independent of the center choice, which would require j to be fixed. But $j^*(k)$ depends on ε (it is defined by the argmin), so cross-terms like $\mathbb{E}[\varepsilon^\top (\mu_k - \mu'_{j^*(k)})]$ do not vanish in general.

This invalidates Lemma 5, which is the cornerstone of the identifiability argument.

3. Major Issues (Need Substantial Work)

Issue 5: Confusion between fixed and scaling $\Delta_{\min}$

The document oscillates between two incompatible interpretations of $\Delta_{\min}$:

- **Interpretation A** (lines 42, 69): $\Delta_{\min}$ is a fixed property of the data-generating process, determined by the true state centers $\mu_1, \dots, \mu_K$.
- **Interpretation B** (lines 486-510, 539): $\Delta_{\min}$ scales with n and K via sphere-packing constraints $\Delta_{\min}^2 = O(K^{-2/d_\phi})$.

The separation condition $\Delta_{\min}^2 \geq C_0 \sigma^2 K \log n / n$ makes sense as a finite-sample sufficient condition under Interpretation A but becomes self-contradictory under Interpretation B (where $\Delta_{\min}$ is determined by K , not n).

The “reconciliation” section (lines 486-510) mixes both interpretations and the algebra is inconsistent. The sphere-packing argument imposes an *upper* bound on $\Delta_{\min}$ given K , while the separation condition imposes a *lower* bound. For large K , these bounds can conflict, preventing any such K from existing.

The derivation of the $1/3$ exponent only works under a very specific set of coupled scaling assumptions that are never stated explicitly.

Issue 6: Lemma 6’s conversion from risk gap to misclassification is not rigorous

Three specific problems:

- (a) The $\pi_k \cdot (\Delta_{\min}/2)^2/2$ factor (line 328) is pulled from thin air. The

author states “accounting for noise” but doesn’t show the calculation. Where does the factor of $1/2$ come from? Why $(\Delta_{\min}/2)^2$ and not $(\Delta_{\min}/2)^2$? This is a critical quantitative claim that directly enters the main theorem’s rate, but it’s unsupported.

(b) The “boundary effects” handling (lines 335-341) is hand-wavy.

The author writes that boundary effects are:

$$O\left(K \cdot \exp\left(-\frac{\Delta_{\min}^2}{8\sigma^2}\right)\right)$$

and claims this is “negligible under the separation condition.” But: - The constant 8 in the denominator is never derived. - The K factor outside the exponential could dominate if K grows quickly. - Under the scaling $\Delta_{\min}^2 = \Theta(K \log n/n)$, the exponential becomes $\exp(-cK \log n/n)$, which is close to 1 when $K \log n/n$ is small, not negligible.

(c) The argument that misclassification only occurs when $\|\hat{\mu}_{\tau(k)} - \mu_k\| \geq \Delta_{\min}/2$ (lines 321-325) ignores cumulative effects. A point from state k could be misclassified even if all centers are well-estimated, if it falls closer to a different estimated center due to noise. The triangle inequality argument is:

$$\begin{aligned}\|\phi(x) - \hat{\mu}_{\tau(k)}\| &\leq \|\mu_k + \varepsilon - \mu_k\| + \|\mu_k - \hat{\mu}_{\tau(k)}\| < \|\varepsilon\| + \frac{\Delta_{\min}}{2} \\ \|\phi(x) - \hat{\mu}_j\| &\geq \|\mu_k + \varepsilon - \mu_j\| - \|\mu_j - \hat{\mu}_j\| \geq \Delta_{\min} - \|\varepsilon\| - \frac{\Delta_{\min}}{2}\end{aligned}$$

The comparison is $\|\varepsilon\| + \frac{\Delta_{\min}}{2}$ vs. $\frac{\Delta_{\min}}{2} - \|\varepsilon\|$, and the inequality $\|\varepsilon\| + \frac{\Delta_{\min}}{2} < \frac{\Delta_{\min}}{2} - \|\varepsilon\|$ requires $\|\varepsilon\| < 0$ (impossible). The argument as stated does not work. The bound needs a more careful treatment of noise.

Issue 7: The covering number bound is insufficient

Lemma 4 bounds the covering number of K -center sets, but the relevant function class for k -means is $f_\mu(x) = \min_k \|x - \mu_k\|_2^2$. The covering number of this function class in the $L_\infty(P_n)$ metric may be larger than the covering number of the center class in the ℓ_∞ metric, because:

- The Lipschitz constant C_1 scales with both M and $\sqrt{\sigma^2 d_\phi}$ (which involves the ambient dimension).
- The function $g_x(\mu) = \min_k \|\phi(x) - \mu_k\|_2^2$ is not Lipschitz with a *uniform* constant across all x – the claim in line 164 that it is L -Lipschitz for all x with the same L requires $\|\phi(x)\|_2$ bounded uniformly, which holds only with high probability, not almost surely.

The author acknowledges that VC dimension would give a similar bound (line 249), but this is misleading. The VC dimension of k -means in $\mathbb{R}^d$ is not $O(Kd)$ – it grows super-linearly in K because the number of possible labelings grows as K^{nd} (the number of Voronoi partitions). A proper VC bound would give $\log \mathcal{N} = \Omega(K \log K)$, not $O(K)$.

Issue 8: Two-layer extension (Corollary 2) assumes the result

Corollary 2 requires $\kappa(W) = O(1)$, which is asserted without proof (line 591). Since W is learned from data – specifically from expert error patterns – there is no guarantee it is well-conditioned. In fact: - If W is estimated from finite data, it has estimation error that degrades the signal-to-noise ratio. - The claim that W can *increase* separation (line 595) is stated without any proof or reference. - The singular values of W depend on the eigenstructure of the cross-covariance between features and expert errors, which is itself estimated. No finite-sample bounds are provided.

Corollary 2 does not prove anything about the two-layer extension. It merely re-states Theorem 3 under the assumption that W is well-conditioned and then asserts (without proof) that this condition holds in practice.

Issue 9: The sub-Gaussian assumption is unverified

The proof assumes ε is sub-Gaussian with parameter σ^2 (line 56). For SCX’s actual features:

- **ACE descriptors:** These are many-body expansion coefficients, typically bounded but with heavy-tailed distributions (some coefficients are near-zero, others can be large). Are they sub-Gaussian? Not obviously.
- **CNN latent features:** These are outputs of ReLU networks, which can have heavy-tailed coordinate distributions. The sub-Gaussian assumption is a strong parametric assumption that is rarely checked in practice.
- **Tabular features:** These are bounded by construction, but can have discrete or categorical components.

If the features are only bounded (not sub-Gaussian), the tail bounds degrade from $\exp(-cn\delta^2)$ to $\exp(-cn\delta)$ (via Hoeffding), which would change the rates entirely.

Issue 10: Appendix A’s sub-Gaussian norm bound is for Gaussians, not sub-Gaussians

Line 770 uses the bound $\|\varepsilon\|_2^2 \leq \sigma^2(d_\phi + 2\sqrt{d_\phi \log n} + 2\log n)$, which is a standard chi-squared concentration bound for *Gaussian* vectors. For sub-

Gaussian vectors, the concentration of the Euclidean norm is different: it depends on the sub-Gaussian norm $K = \sup_{\|u\|=1} \|\langle \varepsilon, u \rangle\|_{\psi_2}$, and the bound is:

$$\|\varepsilon\|_2 \leq CK(\sqrt{d_\phi} + t) \quad \text{with probability } 1 - 2\exp(-t^2)$$

but the constant C depends on the specific sub-Gaussian definition and the covariance structure. The chi-squared-like bound used by the author only holds for *exact* Gaussian vectors, not general sub-Gaussian vectors.

4. Minor Issues

1. **Line 160:** The Lipschitz constant $C_1 = 4(M + \sqrt{\sigma^2 d_\phi})$ uses $\sqrt{\sigma^2 d_\phi}$ which is $\sigma\sqrt{d_\phi}$, but the sub-Gaussian parameter σ^2 has units of variance. This is fine but the notation is confusing: σ appears both as standard deviation and as a sub-Gaussian parameter.
2. **Line 514:** The misclassification rate is written as $O(\sqrt{n/(K \log n)})^{-1}$ which is ambiguous – it could mean $O(1/\sqrt{n/(K \log n)}) = O(\sqrt{K \log n/n})$ OR $O(\sqrt{n/(K \log n)})^{-1}$ meaning the whole thing raised to -1 , which would be $O(\sqrt{K \log n/n})$. Either way, as shown in Issue 1, the rate explodes for $K = o(n^{1/3})$.
3. **Corollary 1:** The formula $K^* \asymp (n/\log M)^{1/3}$ is dimensionally incoherent. M is a bound on center norms (units: feature magnitude). Taking $\log M$ inside the cube root makes no physical sense – if M has units, $\log M$ is not well-defined. This should be $K^* \asymp (n/(\sigma^2 C_0^2))^{1/3}$ or similar, with the dependence on M appearing only in constants.

4. **Line 492:** “sphere-packing bounds” is misspelled as “sphere-packing” (minor typo, but appears consistently).
5. **Line 371:** The simplification shows $Kd_\phi \log n$ as the first term, but this assumes $\log(1 + 8C_1M/\delta_n) \approx \log n$, which is false (see Issue 2).
6. **Line 249:** “The metric entropy bound $O(Kd_\phi \log(1/\varepsilon))$ obtained here is sufficient for our purposes” – this note is dismissive of an important technical point. The covering number for the function class $\{f_\mu\}$ could differ from the covering number for the parameter class $\{\mu\}$, and this difference matters for the constants in the theorem.
7. **Line 104:** The separation condition uses $\log n$, but the standard sub-Gaussian tail bound uses $\log(1/\delta)$. The appearance of $\log n$ rather than $\log K$ or $\log(1/\delta)$ is not justified – it seems to come from a union bound over n samples, but this is never explained.
8. **Line 285:** The author writes “For a fixed true center μ_k , let $j^*(k) = \arg \min_j \|\mu_k - \mu'_j\|_2$ ” – but this ignores that the k -means assignment is based on $\|\phi(x) - \mu'_j\|_2 = \|\mu_k + \varepsilon - \mu'_j\|_2$, not $\|\mu_k - \mu'_j\|_2$. The argmin depends on ε , so fixing j^* independent of ε is incorrect.
9. **Proof of Theorem 3, lines 392-515:** This entire section reads as a working draft, not a finished proof. The author walks through multiple rejected approaches (“Wait – this suggests...”, “Let me re-derive more carefully...”, “Let me re-examine...”, “Corrected scaling analysis...”, “Reconciliation of $K = o(n^{1/3})$ ”). These are signs that the author is reasoning through the problem in real-time, not presenting a finalized proof. A final draft should remove this scaffolding.
10. **Line 404:** “which requires $K \log n = o(n)$, i.e., $K = o(n/\log n)$ ” – but the expression is $\sqrt{n/(K \log n)}$, which goes to 0 when $K \log n = \omega(n)$,

not $o(n)$. The inequality is reversed.

5. Recommended Fixes for Each Issue

Issue	Recommendation
1 (FATAL)	Re-derive the misclassification rate. The bound must be expressed as $O(\sqrt{K \log n/n}/\Delta_{\min}^2)$. If $\Delta_{\min}$ is fixed (does not scale with n), the rate is $O(\sqrt{K \log n/n})$, which gives $K = o(n/\log n)$ for convergence – not $o(n^{1/3})$. If $\Delta_{\min}$ scales, state the scaling explicitly and re-derive all bounds.
2 (FATAL)	Re-do the uniform convergence calculation. The correct choice of δ_n should make the exponent negative. Either increase δ_n (to say $\delta_n = C\sqrt{Kd_\phi(\log n)^2/n}$ to get an extra $\log n$ factor) or use a sharper entropy bound.
3 (FATAL)	This is a structural issue. Two options: (a) Restrict to algorithms that provably find the global k -means minimizer (only possible for small K or special structure). (b) Prove that Lloyd’s algorithm with appropriate initialization finds a solution that is “good enough” – this would require stability of the local minimum (following Rakhlin & Caponnetto, 2007) or a perturbation analysis. Option (c): acknowledge the gap and frame the result as about the oracle.

Issue	Recommendation
4 (FATAL)	Re-do Lemma 5 properly. The key calculation is: $W(\mu) - W(\mu^*) = \sum_k \pi_k \mathbb{E}[\min_j \ \mu_k + \varepsilon - \mu'_j\ ^2 - \ \varepsilon\ ^2]$. The correct lower bound requires a “gap” condition: for any $\mu \neq \mu^*$, there exists at least one k such that $\ \mu_k - \mu'_{\sigma(k)}\ \geq \Delta_{\min}/2$ for any permutation σ, and then one must lower-bound the expectation over ε using sub-Gaussian tail bounds on the assignment probabilities. This is doable but non-trivial.
5	Clarify which interpretation is intended. If $\Delta_{\min}$ is a fixed property of the data, delete all scaling analysis (lines 486-515). If $\Delta_{\min}$ scales, derive this from explicit assumptions about how the centers change as K grows.
6	(a) Derive the constant $1/2$ or replace with a properly bounded expression. (b) Provide an explicit bound on boundary effects using the sub-Gaussian tail, not an $O(\cdot)$ statement. (c) Fix the triangle inequality argument.
7	Either (a) bound the metric entropy of the function class $\{f_\mu\}$ directly, or (b) cite a known VC bound for k-means (e.g., the Natarajan dimension). The note in line 249 suggesting VC dimension gives the same bound is misleading – the VC dimension of k-means is $O(Kd \log(Kd))$ at minimum.

Issue	Recommendation
8	Provide finite-sample bounds on $\kappa(W)$ or drop Corollary 2 until such bounds are available. The current version proves nothing about the two-layer extension.
9	Verify the sub-Gaussian assumption empirically for SCX features, or replace with a boundedness assumption (which gives weaker rates but requires fewer assumptions).
10	Use the correct sub-Gaussian norm concentration bound. The chi-squared bound is only valid for Gaussian vectors.
Structural	Remove the working-draft scaffolding in the proof of Theorem 3 (lines 392-515). A published proof should not contain phrases like “Wait – this suggests...” or “Let me re-derive more carefully.”

6. Summary

The document attempts to prove a non-trivial extension of Pollard’s (1981) k -means consistency to the setting $K \rightarrow \infty$, with explicit rates depending on the separation-to-noise ratio. This is a worthy goal. However, the proof as written contains:

- **2 fatal mathematical errors** that reverse the direction of the main result (the claimed convergence is actually divergence).

REVIEW OF “THEOREM 3: CLUSTER CONSISTENCY OF STATE DISCOVERY UNDER STP

- **1 fatal computational gap** (assuming the global optimum of an NP-hard problem).
- **1 fatal error in Lemma 5**, the identifiability lemma, where the algebra is unjustified.
- **5+ major gaps** in rigor (boundary effects, metric entropy, two-layer extension, sub-Gaussian assumption, confusion between fixed and scaling parameters).

The paper should not be published in its current form. The core idea is interesting but the technical execution does not meet the standard of the Annals of Statistics.

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